UNIVERSITY
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Master's thesis

Topological quantum computing with the Kitaev chain

Solving quantum computing and
providing the world with infinite porridge

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Abstract

Born to dilly dally, forced to be batman

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Chapter 1

Introduction

Blablabla Kitaev is a genius
Joakim Bergli for president

Chapter 2

The Kitaev chain

In this chapter we closely follow the original theory laid out by Kitaev [1], re-deriving the main results and filling in with intermediate steps and discussion for clarity. The Kitaev chain is a toy model of a one dimensional spinless p-wave superconductor, and is an example of material that can realize a topological superconducting phase. It consists of a chain of spinless fermions with nearest-neighbour hopping and superconducting pairing, under the influence of a chemical potential. The Hamiltonian takes the form

$$H = \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] \tag{2.1}$$

where a_j and $a_j^\dagger$ are annihilation and creation operators for the fermionic sites j in the chain. ω is a hopping amplitude, μ is the chemical potential, and $\Delta = |\Delta| e^{i\theta}$ is the induced superconducting gap. The Hamiltonian serves to give rise to Majorana fermion quasiparticles at its boundaries. These Majorana fermions can be used as qubits, and a network of chains can be used for quantum computation.

Majorana fermions are interesting candidates for qubits as their topological nature provides robustness against local perturbations and certain sources of decoherence. Their non-Abelian nature also allows for braiding, where the majorana fermions exchange position and information about the path is stored in the wavefunction.

2.1 Majorana fermions

A Majorana fermion is a particle that is its own antiparticle. The Majorana operators can be defined as

$$c_{2j-1} = a_j + a_j^\dagger, \quad c_{2j} = \frac{a_j - a_j^\dagger}{i}, \quad (j = 1, ..., N) \tag{2.2}$$

which are hermitian, and anticommute, meaning they satisfy the relations

$$c_m^\dagger = c_m, \quad \{c_l, c_m\} = c_l c_m + c_m c_l = 2\delta_{lm}, \quad (l, m = 1, ..., 2N). \tag{2.3}$$

Using the definitions of the majorana operators 2.2 we can find the number operator for a site j

$$a_j^\dagger a_j = \frac{1}{2} \left(1 + i c_{2j-1} c_{2j} \right), \tag{2.4}$$

where the term $ic_{2j-1}c_{2j}$ corresponds to a phase error. If c_{2j-1} and c_{2j} are not on the same site, but separated in space, then $c_{2j-1}c_{2j}$ would represent an interaction between two separated Majorana sites, requiring the environment to couple the Majorana fermions simultaneously to create the phase error. Physical noise and imperfections are typically local, so placing the Majorana sites far enough away from each other would decrease the risk of a phase error. Further, a single Majorana operator cannot appear by itself in a reasonable Hamiltonian as it would not preserve fermionic parity; flipping the system from an even to odd (or vice versa) number of fermions. As such, the Majorana operators will always come in pairs $c_i c_l$, and thus no parity preserving term can involve just one isolated Majorana. Together this tells us that errors would either have to break parity, or couple both separated Majoranas at once, which is highly unlikely and thus the Majorana fermions are protected from local errors.

2.1.1 Rewriting the Hamiltonian in terms of Majorana operators

We can express the Hamiltonian in terms of Majorana operators. As we will see, this transformation will provide insight about the structure of the phases and the emergence of the edge modes. Before doing so we simplify the calculation by hiding the phase θ in the majorana operators

$$c_{2j-1} = e^{i\frac{\theta}{2}}a_j + e^{-i\frac{\theta}{2}}a_j^\dagger \tag{2.5}$$

$$c_{2j} = -ie^{i\frac{\theta}{2}}a_j + ie^{-i\frac{\theta}{2}}a_j^\dagger. \tag{2.6}$$

We can now perform the transformation of the Hamiltonian by inverting these relations and inserting into the original Hamiltonian 2.1. Together with the hermiticity and anticommutation relations 2.3 we get the Hamiltonian

$$H = \frac{i}{2}\sum_j \left[-\mu c_{2j-1}c_{2j} + (\omega + |\Delta|)\, c_{2j}c_{2j+1} + (-\omega + |\Delta|)\, c_{2j-1}c_{2j+2}\right]. \tag{2.7}$$

This calculation is shown in appendix A.1. Staring at this Hamiltonian, we can first see that it is on the general quadratic Majorana form

$$H = \frac{i}{4}\sum_{l,m} A_{lm}c_l c_m, \quad A^T = -A \tag{2.8}$$

with A being a real and antisymmetric matrix, encoding the intrinsic particle-hole symmetry of the superconducting system. The single particle spectrum is fully determined by the eigenvalues of A, reducing the problem from a 2^L-dimensional many-body operator problem in Fock space to a single-particle linear algebra problem on a $2L \times 2L$ matrix. This is the basis of BdG, as we will get back to later. Further, we see that all direct mentions of the complex phase θ are gone from H. This reflects that θ can be removed by an appropriate gauge transformation, and is therefore irrelevant for the bulk spectrum and the topological classification of the phases, which we will also touch upon later.

The Majorana representation also gives us a geometric picture of the chain. The Hamiltonian now consists of three terms, providing three different types of couplings. An on-site term $-\mu c_{2j-1}c_{2j}$, and two nearest-neighbour terms $(\omega + |\Delta|)c_{2j}c_{2j+1}$ and $(-\omega + |\Delta|)c_{2j-1}c_{2j+2}$. The on-site term couples two Majorana modes per site j, so for a chain of length L we get $2L$ Majorana operators. We can garner further insight by studying specific cases for the parameters μ, ω and $|\Delta|$ of the Hamiltonian.

Trivial case $\omega = |\Delta| = 0, \mu > 0$

In the case where μ dominates the Hamiltonian reduces to

$$H = -\mu \sum_j \left(a_j^\dagger a_j - \frac{1}{2} \right) = \frac{i}{2} \mu \sum_j c_{2j-1} c_{2j}. \tag{2.9}$$

In this case the system is topologically trivial as the Majoranas on each site are glued together to form a complex fermion. There are no unpaired Majorana fermions at the edges of the chain, making this case uninteresting for further study.

Topological case $\omega = |\Delta| > 0, \mu = 0$

In the case where the hopping and superconducting parameters dominate the Hamiltonian reduces to

$$H = i \sum_j \omega c_{2j} c_{2j+1}, \tag{2.10}$$

where the nearest neighbours between different sites dominate. This is an interesting case as c_2 couples to c_3, c_4 couples to c_5, and so on. But c_1 and c_{2L}, the Majorana edge states, do not appear in the Hamiltonian. They have however not disappeared physically, but rather remain as zero-energy Majorana modes localized at the edges of the wire. Together they form a nonlocal fermionic zero mode. This is the most important insight from the transformation of the Hamiltonian as these are the states we were looking to create.

We can now define new annihilation and creation operators spanning the sites j and $j+1$

$$d_j = \frac{1}{2} \left(c_{2j} + i c_{2j+1} \right) \tag{2.11}$$

$$d_j^\dagger = \frac{1}{2} \left(c_{2j} - i c_{2j+1} \right). \tag{2.12}$$

Such that we can rewrite the Hamiltonian

$$H = 2\omega \sum_{j=1} \left(f_j^\dagger f_j - \frac{1}{2} \right). \tag{2.13}$$

As the edge states are not mentioned in the Hamiltonian, this expression fully determines the energy spectrum of the bulk. We see that the bulk consists of fermionic modes with energy $\propto 2\omega$. Further, the ground states satisfy the condition $d_j|\psi\rangle = 0$ for all j. As such, all these modes will remain empty, and any ground state $|\psi\rangle$ is fully determined by the edge majoranas c_1 and c_{2L}. Therefore we get two orthogonal states where the above ground state condition for the bulk is true; $|\psi_0\rangle$ and $|\psi_1\rangle$. These ground states will be eigenstates of the hermitian operator $-ic_1c_{2L}$ as the edge Majoranas represent unpaired degrees of freedom. This operator also squares to 1, and as such has eigenvalues ± 1. We get

$$-ic_1c_{2L}|\psi_0\rangle = |\psi_0\rangle, \quad -ic_1c_{2L}|\psi_1\rangle = -|\psi_1\rangle, \tag{2.14}$$

where the labeling of the states were chosen arbitrarily. We see that the total fermion parity operator

$$P = \prod_j (-ic_{2j-1}c_{2j}) \tag{2.15}$$

has the same contribution from the bulk for both ground states. So the difference in parity is solely a result of the edge contribution $-ic_1c_{2L}$. The even parity of $|\psi_0\rangle$ represents the state being a superposition of states with an even number of electrons. And $|\psi_1\rangle$ has an odd parity, representing an odd number of electrons. Together they reflect the boundary degrees of freedom from the unpaired Majorana modes at the edges.

Arbitrary values of ω, μ, Δ

For arbitrary values of ω, μ, Δ the Hamiltonian has the general quadratic form (2.8) in terms of the Majorana operators. This means the Hamiltonian can be diagonalized into independent fermionic modes by a change of basis, implemented by a matrix W. We require that W is a $2N \times 2N$ real orthogonal ($W^TW = WW^T = I$) matrix, and that it has two further properties; the first being that it defines new operators $\{b'_m, b''_m\}$ by the linear transformation $b = Wc$, or

$$\begin{pmatrix} b'_1 \\ b''_1 \\ \vdots \\ b'_N \\ b''_N \end{pmatrix} = W \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_{2N-1} \\ c_{2N} \end{pmatrix}. \tag{2.16}$$

This property is ensured by W being real and orthogonal, such that after the transformation the Majorana anticommutation relations are preserved. This can be shown using $b_\gamma = \sum_\alpha W_{\gamma\alpha}c_\alpha$, where $W_{\gamma\alpha}$ are just real numbers, meaning they commute

$$\{b_\gamma, b_\delta\} = \sum_{\alpha,\beta} W_{\gamma\alpha}W_{\delta\beta}\{c_\alpha, c_\beta\} \tag{2.17}$$

$$= \sum_{\alpha,\beta} W_{\gamma\alpha}W_{\delta\beta}2\delta_{\alpha\beta} \tag{2.18}$$

$$= 2\sum_{\alpha} W_{\gamma\alpha}W_{\delta\alpha} \tag{2.19}$$

$$= 2\sum_{\alpha} W_{\gamma\alpha}W^T_{\alpha\delta} \tag{2.20}$$

$$= 2\delta_{\gamma\delta} \tag{2.21}$$

where the last step is a result of W being orthogonal. The second property stems from the antisymmetry of A, such that in this basis A acquires a canonical block-diagonal form

$$WAW^T = \begin{pmatrix} 0 & \epsilon_1 & & & \\ -\epsilon_1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & \epsilon_N \\ & & & -\epsilon_N & 0 \end{pmatrix}. \tag{2.22}$$

This condition stems from a mathematical theorem on the canonical form of real antisymmetric matrices, which states that if we have a real antisymmetric matrix A of size $2N \times 2N$ there exists a real orthogonal matrix W such that A can be transformed into a block-diagonal form

$$WAW^T = \mathrm{diag}(C_1, C_2, \cdots, C_N), \quad C_j = \begin{pmatrix} 0 & -\lambda_j \\ \lambda_j & 0 \end{pmatrix} \tag{2.23}$$

with $\lambda_j > 0$ being real numbers, stemming from the pairs of complex eigenvalues $\pm i\lambda_j$ for A. This theorem is a standard result in matrix analysis, and a proof can be found in [2] [1]. Once we have a matrix W that satisfies the above conditions we can reduce the Hamiltonian further to a canonical form by inverting the transformation $b_\gamma = \sum_\alpha W_{\gamma\alpha} c_\alpha$ to $c_\alpha = \sum_\gamma W^T_{\alpha\gamma} b_\gamma = \sum_\gamma W_{\gamma\alpha} b_\gamma$

$$
\begin{aligned}
H_{\text{canonical}} &= \frac{i}{4} \sum_{\alpha,\beta} A_{\alpha\beta} c_\alpha c_\beta && (2.24)\\
&= \frac{i}{4} \sum_{\alpha,\beta,\gamma,\delta} A_{\alpha\beta} W_{\gamma\alpha} b_\gamma W_{\delta\beta} b_\delta && (2.25)\\
&= \frac{i}{4} \sum_{\alpha,\beta,\gamma,\delta} W_{\gamma\alpha} A_{\alpha\beta} W_{\delta\beta} b_\gamma b_\delta && (2.26)\\
&= \frac{i}{4} \sum_{\gamma,\delta} \left(W A W^T \right)_{\gamma\delta} b_\gamma b_\delta && (2.27)\\
&= \frac{i}{4} \sum_{m=1}^{N} \begin{pmatrix} b'_m & b''_m \end{pmatrix} \begin{pmatrix} 0 & \epsilon_m \\ -\epsilon_m & 0 \end{pmatrix} \begin{pmatrix} b'_m \\ b''_m \end{pmatrix} && (2.28)\\
&= \frac{i}{4} \sum_{m=1}^{N} \epsilon_m \left(b'_m b''_m - b''_m b'_m \right) && (2.29)\\
&= \frac{i}{4} \sum_{m=1}^{N} \epsilon_m 2 b'_m b''_m && (2.30)\\
&= \frac{i}{2} \sum_{m=1}^{N} \epsilon_m b'_m b''_m = \sum_{m=1}^{N} \epsilon_m \left(\tilde{a}^\dagger_m \tilde{a}_m - \frac{1}{2} \right) \quad (\epsilon_m \geq 0). && (2.31)
\end{aligned}
$$

Where we used that $\{b'_m, b''_m\} = 0$ as they are separate components of b. In this form b'_m and b''_m are real linear combinations of c_{2j-1} and c_{2j}, retaining the commutation relations. N is the number of fermionic sites in the system, here $N = L$. The complex fermion quasiparticles are defined as $\tilde{a}_m = \frac{1}{2}\left(b'_m + i b''_m\right)$ and $\tilde{a}^\dagger_m = \frac{1}{2}\left(b'_m - i b''_m\right)$.

The key take-away here is that the Kitaev chain Hamiltonian, due to being quadratic, can be diagonalized into independent quasi particles $\tilde{a}_m$ with energies ϵ_m. This is how the problem is reduced from a 2^L-dimensional many-body problem to single-particle energies and modes on a $2L \times 2L$ matrix.

2.2 Phases of the Kitaev chain

2.2.1 Phases of the bulk spectrum

The bulk spectrum for energy vs. momentum q for a chain of infinite length is given by [1]

$$
\epsilon(q) = \pm \sqrt{\left(2\omega \cos(q) + \mu\right)^2 + 4|\Delta|^2 \sin^2(q)}, \quad -\pi \leq q \leq \pi. \tag{2.32}
$$

Finding this relation is done by Fourier transforming the Hamiltonian to momentum space as shown in appendix A.2 and then finding its eigenvalues, as we will show in 2.3.

[1] There are apparently many books on this topic, none of which i had access to. This is the best source i could find without proving it directly.

This spectrum will tell us where the system is gapped ($\epsilon(q) \neq 0$), and where gap closings ($\epsilon(q) = 0$) occur. The gap closings mark boundaries between topological phases.

- - - - - - - - - - - - - - - - why? - "In general, topological phases are defined for gapped systems. As long as the bulk gap stays open, one can vary parameters adiabatically without changing the topological class. Hence, gap closings ($\epsilon(q) = 0$ for some q) signal phase transitions and mark the boundaries between distinct topological phases." - Fra chat -

We now find the gap closings

$$\epsilon(q) = 0 \tag{2.33}$$

$$\pm\sqrt{(2\omega\cos(q) + \mu)^2 + 4|\Delta|^2\sin^2(q)} = 0 \tag{2.34}$$

$$(2\omega\cos(q) + \mu)^2 + 4|\Delta|^2\sin^2(q) = 0 \tag{2.35}$$

Both terms in this equation are real values squared, and must therefore be nonnegative. They must therefore both individually vanish to fulfill the equation, and we thus get two equations

$$2\omega\cos(q) + \mu = 0, \quad \sin(q) = 0. \tag{2.36}$$

From the second equation we get $\sin(q) = 0$ for $q = 0, \pi$. Plugging this into the first equation we get

$$\begin{aligned} 2\omega\cos(0) + \mu = 0 \qquad & 2\omega\cos(\pi) + \mu = 0 \\ 2\omega = -\mu \qquad & 2\omega = \mu \end{aligned} \tag{2.37}$$

Meaning the bulk gap closes when $2|\omega| = |\mu|$, defining a phase boundary between the trivial and topological superconducting phases.

From our previous findings we know that the system will be in the topological phase when $\mu < \omega$, which is in the domain $|\mu| < 2|\omega|$. We here expect that if the parameters change without crossing the bulk gap boundary $2|\omega| = |\mu|$ the system should remain in the same phase. Further we need $\Delta \neq 0$ to remain in the superconducting phase. Meaning the phases of the system can be divided into

$2|\omega| > |\mu|, \Delta \neq 0$: Topological superconductor (Majorana edge states)
$2|\omega| > |\mu|, \Delta = 0$: Gapless normal metal (no superconductivity)
$2|\omega| = |\mu|, \Delta \neq 0$: The superconducting bulk gap closes (phase boundary)
$2|\omega| < |\mu|, \Delta \neq 0$: Trivial superconductor (no Majorana edge states)
$2|\omega| < |\mu|, \Delta = 0$: Insulator (gapped, no superconductivity)

2.2.2 Boundary modes

In the previous subsection we expected the phase of the system to remain the same as long as the bulk gap boundary is not crossed. We can check that this is true by studying the boundary modes, as they correspond to eigenvectors of A localized near the ends of the chain, or in other words can confirm whether or not the phase is in the topological phase containing Majorana fermions. Such zero modes should have the form

$$b' = \sum_j \left(\alpha'_+ x_+^j + \alpha'_- x_-^j\right) c_{2j-1} \tag{2.38}$$

$$b'' = \sum_j \left(\alpha''_+ x_+^{-j} + \alpha''_- x_-^{-j}\right) c_{2j} \tag{2.39}$$

- - - - - - - - - - - - - - because? - - - - - - - - - -
with

$$x_{\pm} = \frac{-\mu \pm \sqrt{\mu^2 - 4\omega^2 + 4|\Delta|^2}}{2(\omega + |\Delta|)} \tag{2.40}$$

We have previously confirmed that the Majorana fermions can only exist when either (a) $2|\omega| < |\mu|, \Delta \neq 0$ or (b) $2|\omega| > |\mu|, \Delta \neq 0$. We therefore check these two phases.

(a) $2|\omega| < |\mu|, \Delta \neq 0$

We get

$$|x_+| = \left| \frac{-\mu +}{2(\omega + |\Delta|)} \right| > 1 \tag{2.41}$$

$$|x_-| = \left| \frac{-\mu -}{2(\omega + |\Delta|)} \right| < 1 \tag{2.42}$$

Exponential decay of the overlap / tunneling amplitude

2.2.3 Topological phase diagram

From this we get the plot of topological values in Eivinds(?)

2.3 BdG

2.3.1 BdG form and Nambu basis

We want to express the Hamiltonian in momentum space as

$$H = \frac{1}{2} \sum_q \Psi_q^\dagger \mathcal{H}_{\text{BdG}} \Psi_q, \tag{2.43}$$

using the Nambu basis (spinors)

$$\Psi_q^\dagger = \begin{pmatrix} a_q^\dagger & a_{-q} \end{pmatrix}, \quad \Psi_q = \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix}. \tag{2.44}$$

Where the momentum-space fermioninic annihilation and creation operators $a_q, a_q^\dagger$ are defined by the fourier transformation

$$a_j = \frac{1}{\sqrt{N}} \sum_q e^{iqj} a_q, \quad a_j^\dagger = \frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger. \tag{2.45}$$

Where q runs over the Brillouin zone $[-\pi, \pi]$. We see that $\mathcal{H}_{\text{BdG}}$ has to be a 2×2 matrix. If we look at an arbitrary $\mathcal{H}_{\text{BdG}}$ we can see that after multiplication with the Nambu spinors the Hamiltonian becomes

$$H = \frac{1}{2} \sum_q \begin{pmatrix} a_q^\dagger & a_{-q} \end{pmatrix} \begin{pmatrix} a(q) & b(q) \\ c(q) & d(q) \end{pmatrix} \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix} \tag{2.46}$$

$$= \frac{1}{2} \sum_q \left(a(q) a_q^\dagger a_q + b(q) a_q^\dagger a_{-q}^\dagger + c(q) a_{-q} a_q + d(q) a_{-q} a_{-q}^\dagger \right). \tag{2.47}$$

If this is to match the momentum-space transformed Hamiltonian

$$H = \sum_q \left[-\left(\mu + 2\omega\cos(q)\right) a_q^\dagger a_q + \Delta e^{-iq} a_q a_{-q} + \Delta^* e^{-iq} a_q^\dagger a_{-q}^\dagger \right] \tag{2.48}$$

as found in A.2, we see that we must have

$$a(q) = -\mu - 2\omega\cos(q) \tag{2.49}$$
$$b(q) = \Delta^* e^{-iq} \tag{2.50}$$
$$c(q) = -\Delta e^{-iq} \tag{2.51}$$
$$d(q) = \mu + 2\omega\cos(-q) = \mu + 2\omega\cos(q) \tag{2.52}$$
$$\rightarrow \mathcal{H}_{\mathrm{BdG}} = \begin{pmatrix} -\mu - 2\omega\cos(q) & \Delta^* e^{-iq} \\ -\Delta e^{-iq} & \mu + 2\omega\cos(q) \end{pmatrix}. \tag{2.53}$$

Where we used the anti-commutation relations of the operators and dropped inessential constants. Writing out the entire Hamiltonian we have

$$H = \frac{1}{2} \tag{2.54}$$

2.3.2 Diagonalizing the BdG Hamiltonian

By diagonalizing the BdG Hamiltonian we can find the energy spectrum.

Chapter 3

Quantum computation

The main idea for using Majorana fermions for quantum computation is through a process called braiding. In this process the Majoranas are spatially rotated around eachother, producing a difference in phase. In 1D this is however hard to do: If two Majoranas meet, they simply combine into a normal fermion, allowing the stored information to decohere. Several methods are being discussed to overcome this obstacle, and we will closer study especially one of them; the T-junction. In this junction we will show that one can perform the Pauli X and Z gates (and therefore also Y) but we still need an extra gate, not provided by topologically protected operations, in order to create a general quantum computer.

3.1 Logical qubits from Majorana modes

3.1.1 Degenerate ground states

Recalling the previously discussed Kitaev chain Hamiltonian, and the following definition using the Majorana operators c_m with properties

$$d_j = \frac{1}{2}\left(c_{2j} + ic_{2j+1}\right) \tag{3.1}$$

$$d_j^\dagger = \frac{1}{2}\left(c_{2j} - ic_{2j+1}\right) \tag{3.2}$$

$$c_m^\dagger = c_m, \quad \{c_l, c_m\} = 2\delta_{lm}. \tag{3.3}$$

We recall that in the trivial phase, Majoranas on all sites pair up into ordinary fermions with non-zero energy. However in the topological phase the chain experiences unpaired Majoranas at the edges, which we want well-separated (meaning their wavefunctions overlap negligibly). For a 1D wire of $2M$ such edge Majoranas one can generally construct M fermion operators f_{lm} [1] with the properties

$$f_{lm} = \frac{1}{2}\left(c_l + ic_m\right) \tag{3.4}$$

$$\{f_{lm}, f_{lm}^\dagger\} = 1 \tag{3.5}$$

$$f_{ij}^2 = \left(f_{ij}^\dagger\right)^2 = 0 \tag{3.6}$$

These correspond to zero-energy modes, and can give rise to 2^M degenerate ground states which can be labeled by occupation numbers $n_j = f_j^\dagger f_j = 0, 1$.

[1] Note here that f_{lm} is a special case of the the general d_j shown above, only f_{lm} is specifically for the Majorana edge modes

3.1.2 Logical basis states

Four Majoranas - a proper qubit

If we have a single Kitaev chain in the topological phase we can produce 2 Majorana zero mdoes.

Conservation of global fermion parity P - Fixed P means that no pair can encode a qubit

In the case of having two wires in the topological phase we can produce 4 Majorana zero modes, which can be used to create logical basis states

$$|0_L\rangle = |0_{12}, 0_{34}\rangle \tag{3.7}$$

$$|1_L\rangle = |1_{12}, 1_{34}\rangle = f_{12}^\dagger f_{34}^\dagger |0_L\rangle \tag{3.8}$$

$0_L = 0, 0$

Pauli operators

3.1.3 Nonlocality and wavefunction picture

Overlap decays as exp(-L / xi)

[3] [4]

3.2 Braiding and non-abelian statistics (exchange of Majoranas)

3.2.1 General result

Berry phase

Tegning av braiding?

3.2.2 Clifford group

Exchange operator R [5]

X, Z, Y = iXZ

3.2.3 What's missing

We can get X (majoranas 1 and 2 switched), Z (majoranas 2 and 3 switched), and Y = i XZ operations, but missing something like T (pi / 8 rotation)

Something something non-topological

supplementary operations

3.3 Methods using the Kitaev chain

3.3.1 T-junction

The T-junction is a proposed architecture for allowing braiding with a 1D Kitaev chain. Simply put, the idea stems from connecting a 1D chain to the middle of another, producing a T-shape. In this geometry, one can in theory braid the Majoranas by altering the parameters of the Hamiltonian to rotate the edge states around eachother without touching.

Figure 3.1

Exchange of two Majoranas

Pali x y z gates

Double switches = rotations? (Drawing from meeting with Joakim)

Network of wires

3.3.2 Quantum dots

3.3.3 Comparison of T-junction and Quantum dots

3.3.4 Others

3.4 Topological protection

Chapter 4

Numerical methods

In this paper we have numerically simulated the Kitaev chain T-junction using the full Fock space. We have chosen to do so using the adiabatic approximation in hopes of reducing the otherwise overwhelming demand for computational power, aiming to truncate the state space while preserving numerical accuracy. For integration we have used RK4 as it is relatively simple and produces accurate results, although it is not symplectic meaning it can drift from the true result if the time step is not sufficiently small.

4.1 Theory

4.1.1 Hilbert space and many-body basis

Considering a Kitaev chain with L sites we can use an occupation number basis

$$\{|n\rangle\}, \quad |n\rangle = |n_1, n_2, ..., n_L\rangle, \quad n_i \in \{0, 1\}. \tag{4.1}$$

Where 0 represents a site being empty, and 1 represents a site being occupied. This defines our Hilbert space, with a corresponding Hamiltonian $\mathcal{H}$. This allows us to produce a Hamiltonian matrix

$$H_{nm} = \langle n|\mathcal{H}|m\rangle. \tag{4.2}$$

4.1.2 Diagonalizing the Hamiltonian

Diagonalizing by the use of a unitary matrix U with elements c_{nk}

$$H = UDU^{-1} \tag{4.3}$$

where D is diagonal containing the eigenvalues E_k. The energy eigenstates are related to the basis by

$$|\psi_k\rangle = \sum_n c_{nk}|n\rangle. \tag{4.4}$$

We can expand any state in the energy eigenbasis

$$|\phi\rangle = \sum_k a_k|\psi_k\rangle. \tag{4.5}$$

For a time-independent H, the time evolution of U can be written

$$U(t) = e^{-iHt} \tag{4.6}$$

such that

$$\begin{aligned} |\phi(t)\rangle &= U(t)|\phi\rangle && (4.7)\\ &= e^{-iHt}|\phi(0)\rangle && (4.8)\\ &= \sum_k a_k(0)e^{-iHt}|\psi_k\rangle && (4.9)\\ &= \sum_k a_k(t)|\psi_k\rangle && (4.10) \end{aligned}$$

Where $|\phi(0)\rangle = |\phi\rangle, a_k(0) = a_k$ from earlier. We see that the coefficients just pick up time-dependent phases

$$a_k(t) = a_k(0)e^{-iHt} \tag{4.11}$$

Time-dependent Hamiltonian & adiabatic basis

We now want to define a time dependent Hamiltonian with an orthonormal instantaneous basis

$$\begin{aligned} H(t); \quad \{|\psi_k(t)\rangle\} && (4.12)\\ H(t)|\psi_k(t)\rangle = E_k(t)|\psi_k(t)\rangle. && (4.13)\\ && (4.14) \end{aligned}$$

Expanding the full state as

$$|\phi(t)\rangle = \sum_k a_k(t)e^{-i\int_0^t dt'\lambda_k(t')}|\psi_k(t)\rangle \tag{4.15}$$

we can now determine the time evolution of the system using Schrödingers equation

$$\begin{aligned} -iH|\phi\rangle &= \frac{d}{dt}|\phi\rangle && (4.16)\\ -iH\sum_k a_k e^{-i\int_0^t \lambda_k}|\psi_k\rangle &= \frac{d}{dt}\sum_k a_k(t)e^{-i\int_0^t dt'\lambda_k(t')}|\psi_k(t)\rangle && (4.17)\\ -i\sum_k a_k e^{-i\int_0^t \lambda_k}E_k|\psi_k\rangle &= \sum_k \left[\frac{da_k}{dt}e^{-i\int_0^t \lambda_k}|\psi_k\rangle + a_k(-i\lambda_k)e^{-i\int_0^t \lambda_k}|\psi_k\rangle + a_k e^{-i\int_0^t \lambda_k}\frac{d|\psi_k\rangle}{dt}\right] && (4.18) \end{aligned}$$

$$\begin{aligned} -i\sum_k e^{-i\int_0^t \lambda_k}\left[a_k E_k|\psi_k\rangle - i\frac{da_k}{dt}|\psi_k\rangle - a_k\lambda_k|\psi_k\rangle - ia_k\frac{d|\psi_k\rangle}{dt}\right] &= 0 && (4.19)\\ \sum_k e^{-i\int_0^t \lambda_k}\left[a_k\left((E_k-\lambda_k)|\psi_k\rangle - i\frac{d|\psi_k\rangle}{dt}\right) - i\frac{da_k}{dt}|\psi_k\rangle\right] &= 0 && (4.20) \end{aligned}$$

We now project onto $\langle\psi_l|$ as we by orthonormality have $\langle\psi_l|\psi_k\rangle = \delta_{lk}$.

$$\langle\psi_l| \sum_k e^{-i\int_0^t \lambda_k} \left[a_k \left((E_k - \lambda_k)|\psi_k\rangle - i\frac{d|\psi_k\rangle}{dt} \right) - i\frac{da_k}{dt}|\psi_k\rangle \right] = 0 \tag{4.21}$$

$$\sum_k e^{-i\int_0^t \lambda_k} \left[a_k \left((E_k - \lambda_k)\delta_{l,k} - i\langle\psi_l|\frac{d}{dt}|\psi_k\rangle \right) - i\frac{da_k}{dt}\delta_{l,k} \right] = 0 \tag{4.22}$$

$$e^{-i\int_0^t \lambda_l} \left[a_l(E_l - \lambda_l) - i\frac{da_l}{dt} \right] - i\sum_k a_k e^{-i\int_0^t \lambda_k} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle = 0 \tag{4.23}$$

$$a_l(E_l - \lambda_l) - i\frac{da_l}{dt} - ie^{i\int_0^t \lambda_l} \sum_k a_k e^{-i\int_0^t \lambda_k} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle = 0 \tag{4.24}$$

$$\sum_k a_k e^{-i\int_0^t (\lambda_l - \lambda_k)} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle - ia_l(E_l - \lambda_l) = \frac{da_l}{dt} \tag{4.25}$$

Remembering that λ is just a definition used to split off a phase from a_k, we can choose it to help simplify our equation. We see that by choosing

$$\lambda_k = E_k - i\langle\psi_k|\frac{d}{dt}|\psi_k\rangle \tag{4.26}$$

we can remove all diagonal terms.

$$\sum_k a_k e^{-i\int_0^t (\lambda_l - \lambda_k)} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle - ia_l(E_l - E_l - i\langle\psi_l|\frac{d}{dt}|\psi_l\rangle) = \frac{da_l}{dt} \tag{4.27}$$

$$\sum_{k\neq l} a_k e^{-i\int_0^t (\lambda_l - \lambda_k)} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle + a_l e^0 \langle\psi_l|\frac{d}{dt}|\psi_l\rangle - a_l\langle\psi_l|\frac{d}{dt}|\psi_l\rangle = \frac{da_l}{dt} \tag{4.28}$$

$$\sum_{k\neq l} a_k e^{-i\int_0^t (\lambda_l - \lambda_k)} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle = \frac{da_l}{dt} \tag{4.29}$$

4.1.3 RK4 integration

Accuracy

4.1.4 Code structure

I used UiOGPT to help write and debug the code.

4.1.5 Setting up the Kitaev chain Hamiltonian matrix

Using original Hamiltonian 2.1.

Open boundary conditions
Hilbert space basis
Making H_{nm}
For mu term: Use $\sum_j n_j|n\rangle = \sum_j a_j^\dagger a_j|n\rangle = N|n\rangle$
For omega and delta terms: $c_j^\dagger c_{j+1}|n\rangle = c_1 c_2|m_2\rangle$
Use time dependent μ for time dependency.

4.1.6 Finding eigenvectors and eigenvalues by diagonalization

numpy.linalg.eigh

4.1.7 Calculating $\langle\psi_l|\frac{d}{dt}|\psi_k\rangle$ & λ

Finite difference (Numerical derivation) for the bulk $n \in [1, N-1]$

$$\langle\psi_l(t_n)|\frac{d}{dt}|\psi_k(t_n)\rangle = \frac{\langle\psi_l(t_n)|\psi_k(t_{n+1})\rangle - \langle\psi_l(t_n)|\psi_k(t_{n-1})\rangle}{2\Delta t}. \tag{4.30}$$

And a forward difference for $n = 0$

$$\langle\psi_l(t_0)|\frac{d}{dt}|\psi_k(t_0)\rangle = \frac{d}{dt}\left[\langle\psi_l(t_0)|\psi_k(t)\rangle\right]_{t_0} \tag{4.31}$$

$$= \frac{\langle\psi_l(t_0)|\psi_k(t_1)\rangle - \langle\psi_l(t_0)|\psi_k(t_0)\rangle}{t_1 - t_0} \tag{4.32}$$

$$= \frac{\langle\psi_l(t_0)|\psi_k(t_1)\rangle - \delta_{lk}}{dt}. \tag{4.33}$$

And a backward difference to find $n = N - 1$.

4.1.8 Time evolution of amplitudes

Using Euler.
Upgrade to RK4 later.

Evolving the fixed basis

Using H(t) with $\mu(t), \omega(t), \Delta(t)$

Here one can also evolve the ... basis as seen in Eivinds master

Chapter 5

Results & discussion

Eivind the GOAT

What do plots in the adiabatic basis actually tell us?

5.1 Splitting the chain

A previous masters student also did this. Boom i replicated his results life is good

eivind cut means that it's not the central bond thats cut, but rather one out to the side. This matches Eivinds new code.

Adiabatic evolution produces correct adiabatic basis coefficients for just low truncation (trunc = 10) and high dt (dt = 0.1). But the energy spectrum is completely off. Even using trunc = 250 makes the energy spectrum seemingly wrong for the Pauli X-gate (6-8 ground states where 4 is expected and weird bands (a single energy occupying a band, which becomes a single line as trunc increases)). Maybe run a proper test for cutting of a chain?

Schrodinger evolution vs adiabatic evolution

Time and accuracy wrt truncation

5.2 T-junction

I need a supercomputer :/

5.2.1 Pali X-gate

Exchange 2 and 3

5.2.2 Pauli Z-gate

Exchange 1 and 2 (or 3 and 4)

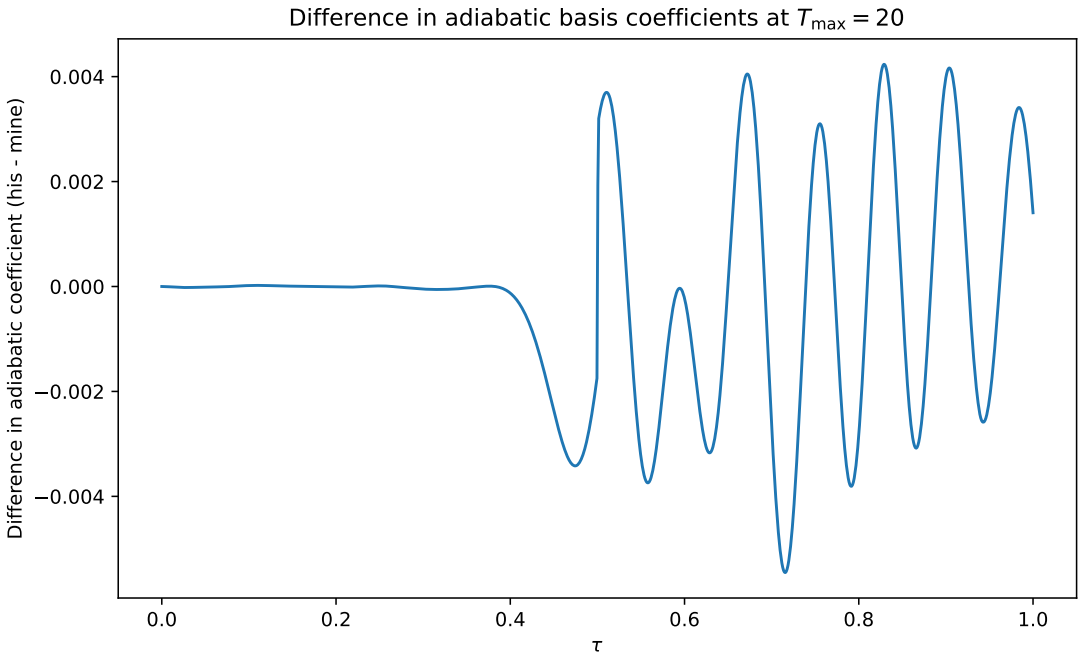


Figure 5.1

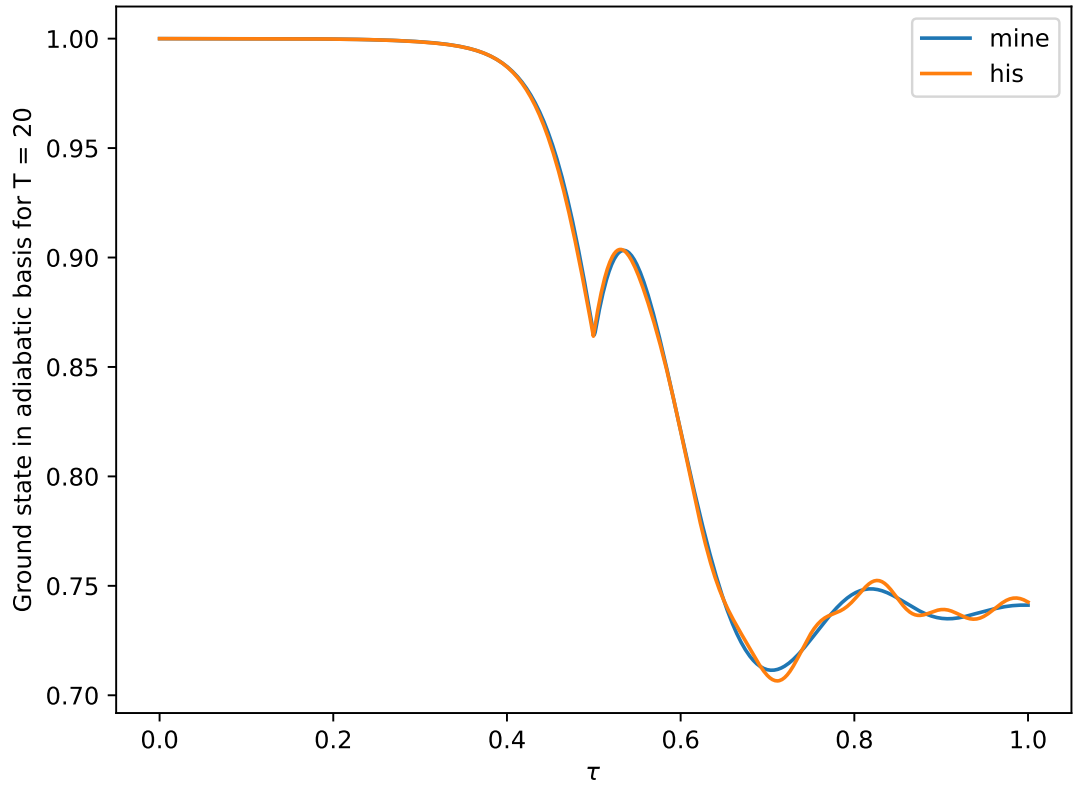


Figure 5.2: $\mathrm{trunct} = 10$

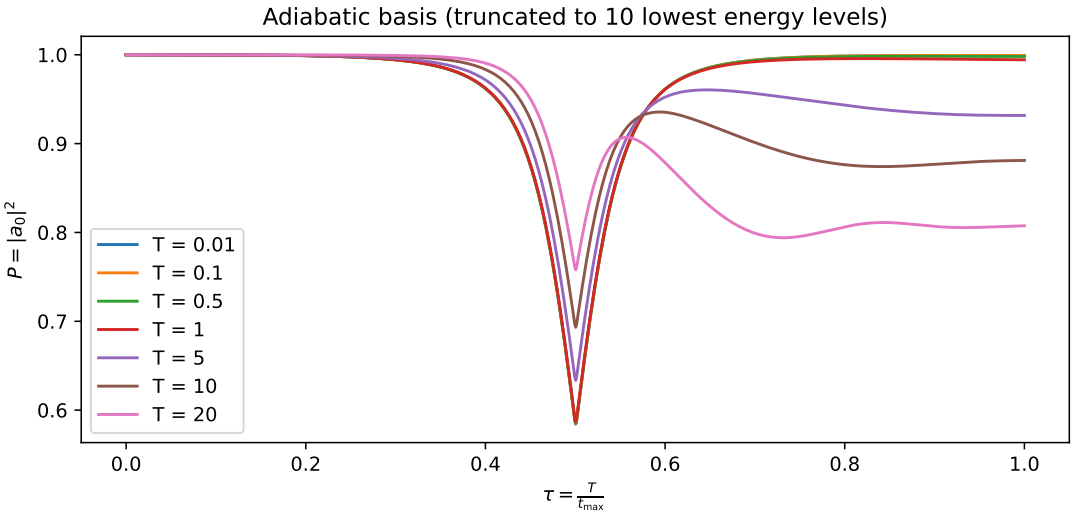


Figure 5.3

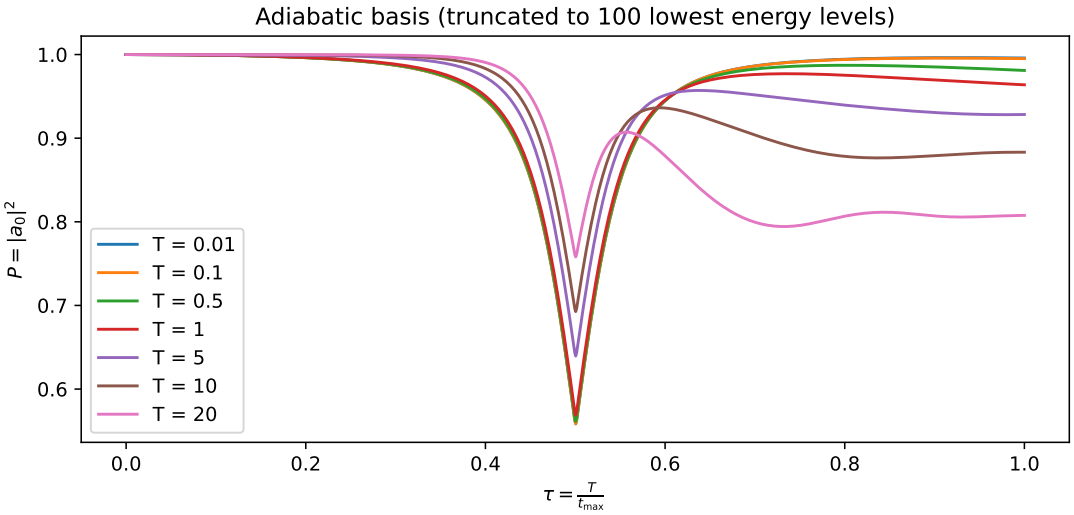


Figure 5.4

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Appendix A

Calculations

A.1 Transforming the Kitaev chain Hamiltonian to Majorana operators

This appendix shows how transform the original Kitaev chain Hamiltonian expressed in fermionic operators a_j and $a_j^\dagger$ to being expressed in terms of the majorana operators c_{2j-1} and $c_2 j$.

Starting with the majorana operators

$$c_{2j-1} = e^{i\frac{\theta}{2}} a_j + e^{-i\frac{\theta}{2}} a_j^\dagger \tag{A.1}$$
$$c_{2j} = -ie^{i\frac{\theta}{2}} a_j + ie^{-i\frac{\theta}{2}} a_j^\dagger. \tag{A.2}$$

We can invert them, leading to four expressions

$$a_j = e^{-i\frac{\theta}{2}} c_{2j-1} - e^{-i\theta} a_j^\dagger \tag{A.3}$$
$$= ie^{-i\frac{\theta}{2}} c_{2j} + e^{-i\theta} a_j^\dagger \tag{A.4}$$
$$a_j^\dagger = e^{i\frac{\theta}{2}} c_{2j-1} - e^{i\theta} a_j \tag{A.5}$$
$$= -ie^{i\frac{\theta}{2}} c_{2j} + e^{i\theta} a_j. \tag{A.6}$$

Combining the expressions for a_j, and for $a_j^\dagger$ we find

$$a_j = \frac{1}{2}\left(e^{-i\frac{\theta}{2}} c_{2j-1} - e^{-i\theta} a_j^\dagger + e^{-i\theta} a_j^\dagger + ie^{-i\frac{\theta}{2}} c_2 j\right) \tag{A.7}$$
$$= \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + ic_{2j}\right) \tag{A.8}$$
$$a_j^\dagger = \frac{1}{2}\left(e^{i\frac{\theta}{2}} c_{2j-1} - e^{i\theta} a_j + e^{i\theta} a_j - ie^{i\frac{\theta}{2}} c_{2j}\right) \tag{A.9}$$
$$= \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - ic_{2j}\right). \tag{A.10}$$

We now substitute these expressions into the original Hamiltonian

$$H = \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] \tag{A.11}$$

$$= -\omega \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2(j+1)-1} + i c_{2(j+1)} \right) \tag{A.12}$$

$$- \omega \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2(j+1)-1} - i c_{2(j+1)} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) \tag{A.13}$$

$$- \mu \sum_j \left(\frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) - \frac{1}{2} \right) \tag{A.14}$$

$$+ \Delta \sum_j \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2(j+1)-1} + i c_{2(j+1)} \right) \tag{A.15}$$

$$+ \Delta^* \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2(j+1)-1} - i c_{2(j+1)} \right) \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \tag{A.16}$$

$$= -\frac{\omega}{4} \sum_j \left(c_{2j-1} c_{2j+1} + i c_{2j-1} c_{2j+2} - i c_{2j} c_{2j+1} + c_{2j} c_{2j+2} \right) \tag{A.17}$$

$$- \frac{\omega}{4} \sum_j \left(c_{2j+1} c_{2j-1} + i c_{2j+1} c_{2j} - i c_{2j+2} c_{2j-1} + c_{2j+2} c_{2j} \right) \tag{A.18}$$

$$- \frac{\mu}{4} \sum_j \left(c_{2j-1}^2 + i c_{2j-1} c_{2j} - i c_{2j} c_{2j-1} + c_{2j}^2 - 2 \right) \tag{A.19}$$

$$+ \frac{|\Delta|}{4} e^{i\theta} \sum_j e^{-i\theta} \left(c_{2j-1} c_{2j+1} + i c_{2j-1} c_{2j+2} + i c_{2j} c_{2j+1} - c_{2j} c_{2j+2} \right) \tag{A.20}$$

$$+ \frac{|\Delta|}{4} e^{-i\theta} \sum_j e^{i\theta} \left(c_{2j+1} c_{2j-1} - i c_{2j+1} c_{2j} - i c_{2j+2} c_{2j-1} - c_{2j+2} c_{2j} \right) \tag{A.21}$$

$$= -\frac{\omega}{4} \sum_j \left(0 + 2 i c_{2j-1} c_{2j+2} - 2 i c_{2j} c_{2j+1} + 0 \right) \tag{A.22}$$

$$- \frac{\mu}{4} \sum_j \left(2 i c_{2j-1} c_{2j} + 2 - 2 \right) \tag{A.23}$$

$$+ \frac{|\Delta|}{4} \sum_j \left(0 + 2 i c_{2j-1} c_{2j+2} + 2 i c_{2j} c_{2j+1} + 0 \right) \tag{A.24}$$

$$= \frac{i}{2} \sum_j \left[-\mu c_{2j-1} c_{2j} + \left(\omega + |\Delta| \right) c_{2j} c_{2j+1} + \left(-\omega + |\Delta| \right) c_{2j-1} c_{2j+2} \right] \tag{A.25}$$

Where we used $\Delta = |\Delta| e^{i\theta}$ and the anticommutation relations

$$\{ c_l, c_m \} = c_l c_m + c_m c_l = 2\delta_{lm} \tag{A.26}$$

$$c_l c_m = -c_m c_l (m \neq l) \tag{A.27}$$

$$c_l^2 = 1. \tag{A.28}$$

A.2 Transforming the Kitaev chain Hamiltonian to momentum space

Taking inspiration from [4] we start from the Kitaev chain Hamiltonian 2.1. Imposing periodic boundary conditions we can Fourier transform the Hamiltonian to momentum

space q. We recall the Fourier transform

$$a_j = \frac{1}{\sqrt{N}} \sum_q e^{iqj} a_q, \quad a_j^\dagger = \frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger \tag{A.29}$$

where q runs over the Brillouin zone $[-\pi, \pi]$. Using this we can transform each term in the Hamiltonian individually for simplicity. Starting with the on-site term

$$H_\mu = -\mu \sum_j \left(a_j^\dagger a_j - \frac{1}{2} \right) \tag{A.30}$$

$$= -\mu \sum_j \left(\left(\frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{iq'j} a_{q'} \right) - \frac{1}{2} \right) \tag{A.31}$$

$$= -\mu \frac{1}{N} \sum_{jqq'} e^{i(q'-q)j} a_q^\dagger a_{q'} + \frac{\mu N}{2} \tag{A.32}$$

$$= -\frac{\mu}{N} \sum_{qq'} N \delta_{qq'} a_q^\dagger a_{q'} + \frac{\mu N}{2} \tag{A.33}$$

$$= -\mu \sum_q a_q^\dagger a_q + \frac{\mu N}{2}. \tag{A.34}$$

Where we used a property of the periodic boundary conditions $\sum_j e^{i(q'-q)j} = N\delta_{qq'}$. Continuing with the hopping term

$$H_\omega = -\omega \sum_j \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) \tag{A.35}$$

where we again split the calculation of the terms; the first term becomes

$$H_\omega^1 = -\omega \sum_j \left(\frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{iq'(j+1)} a_{q'} \right) \tag{A.36}$$

$$= -\frac{\omega}{N} \sum_{jqq'} e^{i(q'-q)j} e^{iq'} a_q^\dagger a_{q'} \tag{A.37}$$

$$= -\frac{\omega}{N} \sum_{qq'} N \delta_{qq'} e^{iq'} a_q^\dagger a_{q'} \tag{A.38}$$

$$= -\omega \sum_q e^{iq} a_q^\dagger a_q. \tag{A.39}$$

And the second term similarly becomes

$$H_\omega^2 = -\omega \sum_q e^{-iq} a_q^\dagger a_q. \tag{A.40}$$

Combining the hopping terms we get

$$H_\omega = H_\omega^1 + H_\omega^2 \tag{A.41}$$

$$= -\omega \sum_q \left(e^{iq} + e^{-iq} \right) a_q^\dagger a_q \tag{A.42}$$

$$= -2\omega \sum_q \cos(q) a_q^\dagger a_q. \tag{A.43}$$

We now consider the pairing term

$$H_\Delta = \sum_j \left(\Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger\right) \tag{A.44}$$

$$\tag{A.45}$$

where we again split the calculation for simplicity

$$H_\Delta^1 = \Delta \sum_j \left(\frac{1}{\sqrt{N}} \sum_q e^{iqj} a_q\right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{iq'(j+1)} a_{q'}\right) \tag{A.46}$$

$$= \frac{\Delta}{N} \sum_{jqq'} e^{i(q+q')j} e^{iq'} a_q a_{q'} \tag{A.47}$$

$$= \frac{\Delta}{N} \sum_{qq'} N\delta_{q(-q')} e^{iq'} a_q a_{q'} \tag{A.48}$$

$$= \Delta \sum_q e^{-iq} a_q a_{-q}. \tag{A.49}$$

And the second term similarly becomes

$$H_\Delta^2 = \Delta^* \left(\frac{1}{\sqrt{N}} \sum_q e^{-iq(j+1)} a_q^\dagger\right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{-iq'j} a_{q'}^\dagger\right) \tag{A.50}$$

$$= \frac{\Delta^*}{N} \sum_{jqq'} e^{-i(q+q')j} e^{-iq} a_q^\dagger a_{q'}^\dagger \tag{A.51}$$

$$= \frac{\Delta^*}{N} \sum_{qq'} N\delta_{q(-q')} e^{-iq} a_q^\dagger a_{q'}^\dagger \tag{A.52}$$

$$= \Delta^* \sum_q e^{-iq} a_q^\dagger a_{-q}^\dagger. \tag{A.53}$$

Combining all terms we get the momentum-space Hamiltonian

$$H = \sum_q \left[-\mu\left(a_q^\dagger a_q\right) - 2\omega\cos(q) a_q^\dagger a_q + \Delta e^{-iq} a_q a_{-q} + \Delta^* e^{-iq} a_q^\dagger a_{-q}^\dagger\right] + \frac{\mu N}{2} \tag{A.54}$$

$$= \sum_q \left[-\left(\mu + 2\omega\cos(q)\right) a_q^\dagger a_q + \Delta e^{-iq} a_q a_{-q} + \Delta^* e^{-iq} a_q^\dagger a_{-q}^\dagger\right] \tag{A.55}$$

Where the last term $+\frac{\mu N}{2}$ can be ignored as it does not affect the spectrum, eigenstates, or any observable that depends on the energy differences.

A.3 Finding the evolution of the system under the adiabatic assumption

Move huge ugly calculation from methods to here